

S-SPARC Enhancement: Measuring & Mitigating AI Over-Reliance in Programming Education

A Telemetry-Driven Behavioral Metric with Productive Friction in E-STRANGE

EXECUTIVE SUMMARY

This proposal outlines an enhancement to the S-SPARC AI assistant and E-STRANGE auto-grader. While generative AI enables rapid code generation, unconstrained access frequently causes cognitive offloading and impairs students' independent debugging skills. We introduce the **AI Over-Reliance Index (ORI)**: an automated, real-time behavioral metric calculated from server telemetry (prompt entropy, problem completeness, conceptual inquiry ratio, and reflection pause adherence) to detect over-reliance and trigger adaptive scaffolding.

1. The Problem: Cognitive Offloading in Programming Labs

In introductory and intermediate programming courses, students frequently fall into a passive interaction loop: copying problem statements directly into conversational AI, copying generated code, and submitting it to automated grading platforms. While assignment scores appear high, students struggle during unassisted laboratory exams when AI tools are unavailable [2], [3].

- **Critical Thinking Decay:** Literature demonstrates that excessive reliance on conversational AI degrades independent problem decomposition and self-debugging abilities [2].
- **Weak Conceptual Transfer:** Instant AI assistance shows minimal correlation with genuine conceptual mastery and long-term retention [3].
- **Impact on Course Outcomes:** High LLM dependency correlates negatively with summative unassisted exam scores [4].
- **Superficial Engagement:** Naturalistic interaction analyses reveal that students rarely reflect critically on generated code snippets before submitting them [5].

2. State of the Art & Research Gap

Existing pedagogical approaches to generative AI in computing education suffer from significant operational limitations:

Current Approach	Main Limitation	S-SPARC Enhancement
Surveys & Questionnaires [2]	Subjective, prone to retrospective recall bias, cannot capture live behavior.	Continuous Telemetry: Automated, real-time logging of prompt structure, dwell times, and cadence.
AI Plagiarism Checkers	High false-positive rate, purely punitive, provides zero learning feedback.	Cognitive Profiling: Maps student problem-solving habits and scaffolding needs over time [5].
Unregulated Commercial Chatbots [5]	Encourages rapid prompt spamming and copy-paste without cognitive effort.	Productive Friction: Enforced 60s reflection pause and stepped Socratic hints [6], [7].

3. Metric Formulation: AI Over-Reliance Index (ORI)

The ORI integrates four unobtrusive telemetry indicators into a single normalized composite score ranging from 0.0 (independent problem solver) to 1.0 (heavily over-reliant):

$$\text{ORI} = 1 - (w_1 Q_{\text{prompt}} + w_2 S_{\text{CIOE}} + w_3 R_{\text{concept}} + w_4 F_{\text{reflect}})$$

All indicators normalized in [0, 1] • Baseline weights: $w_1 = 0.30$, $w_2 = 0.25$, $w_3 = 0.20$, $w_4 = 0.25$

1. Prompt Entropy (Q_{prompt})

$H(X) = -\sum p(x) \log_2 p(x)$ [1]. Measures technical vocabulary richness in queries. Penalizes vague or low-effort prompts ("fix my code").

2. Context Completeness (S_{CIOE})

Evaluates 4 structural dimensions: Context, Input specs, Expected Output, and Error Trace. Rewards well-formulated technical queries.

3. Conceptual Ratio (R_{concept})

Measures the proportion of queries seeking conceptual explanation or algorithm logic (Bloom C2-C4) vs requests for instant code generation (Bloom C1).

4. Reflection Adherence (F_{reflect})

Tracks compliance with the 60s reflection pause [6] alongside keystroke cadence analysis to detect and penalize immediate copy-pasting.

4. Statistical Framework & Variable Mapping (What is Analyzed)

The empirical validation links real-time telemetry indicators with unassisted student learning outcomes through an explicit variable mapping:

Variable Category	Specific Variables	Operational Definition & Measurement Role
Independent Variables (X)	X_1 : Q_{prompt} (Entropy) X_2 : S_{CIOE} (Completeness) X_3 : R_{concept} (Conceptual Ratio) X_4 : F_{reflect} (Friction Adherence)	Continuous telemetry metrics logged per student during programming tasks in E-STRANGE. Evaluates query richness, problem formulation, cognitive inquiry level, and reflection habits.
Dependent Variables (Y)	Y_1 : Unassisted Exam Score Y_2 : GenAI Reliance Scale [3]	Y_1 (Ground Truth Mastery) : Summative programming test score conducted without AI or internet access. Y_2 (Convergent Construct Validity) : Standardized survey score from Hou et al. (2025).
Covariate / Control (Z)	Z: Baseline Prior Ability	Pre-test programming score or cumulative GPA. Used to control for prior competence and estimate the $\text{Entropy} \times \text{Prior Ability}$ interaction effect.

Empirical Weight Calibration Model:

$$Y_{\text{unassisted}} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 (X_1 \times Z) + \epsilon$$

Component weights derived from standardized coefficients: $w_i = |\beta_i| / \sum |\beta_j|$ • Multicollinearity verified via Variance Inflation Factor (VIF < 5.0)

5. Step-by-Step Statistical Testing & Psychometric Procedures

1. Latent Factor Dimensionality (EFA & CFA):

- **Phase 1 (Cohort 1, $N = 35\text{--}50$):** Exploratory Factor Analysis (EFA) using Principal Axis Factoring with Promax rotation to prove the 4 indicators load onto a single latent construct ($KMO > 0.70$, Bartlett's test $p < 0.05$).
- **Phase 2 (Cohort 2, $N \geq 80$):** Independent Confirmatory Factor Analysis (CFA) to confirm construct unidimensionality (Fit criteria: $CFI \geq 0.95$, $TLI \geq 0.95$, $RMSEA \leq 0.06$).

2. **Anti-Circularity 5-Fold Cross-Validation:** To prevent over-optimistic evaluation from fitting on the same dataset, data is partitioned into 5 folds: regression weights are calibrated on 4 training folds and validated on the held-out 5th fold. Cross-validated R^2_{CV} and RMSE are reported.

3. **Convergent Construct Validation:** Pearson and Spearman rank correlation coefficients are evaluated between telemetry-based ORI scores and the standardized GenAI Reliance Scale (Hou et al. 2025 [3]), targeting $r \geq 0.60$ ($p < 0.001$).

4. **Secondary Hypotheses & Family-Wise Error Control:** Evaluated using Linear Mixed-Effects Models (LMM) with Student ID as random effect and Assignment Module as fixed effect. Significance threshold is Bonferroni-adjusted: $\alpha_{\text{adjusted}} = 0.05 / 4 = 0.0125$.

6. Telemetry Database Schema (MySQL Backend)

Interaction events are captured in real-time through the existing S-SPARC and E-STRANGE logging architecture:

Database Field	Data Definition & Granularity	Construct Mapping
<code>prompt_raw_text</code>	UTF-8 query string with millisecond timestamp	Query formulation quality
<code>entropy_score</code>	Shannon Information Entropy $H(X)$ calculated over token stream	Lexical richness (Q_{prompt})
<code>cioe_bitmask</code>	4-bit vector for Context, Input, Output, and Error stack trace	Problem completeness (S_{CIOE})
<code>keystroke_dwell</code>	Inter-keystroke timing variance and anti-paste dwell metrics	Friction integrity (F_{reflect})
<code>execution_metrics</code>	Unit test pass rate, runtime errors, and compilation attempts	Objective code mastery
<code>carbon_footprint</code>	Estimated inference token energy consumption and CO_2 (g)	Compute sustainability

7. Methodological Defenses & Threats to Validity

- **Reactivity & Maturation Confound:** The study employs an A-B design (Modules 1-2 baseline without friction, Modules 3-4 with active friction). Maturation effects over time are transparently disclosed as a study limitation; where multiple lab sections exist, counterbalancing (A-B vs B-A) is applied.
- **Entropy Inversion Paradox:** Advanced students naturally write concise queries. We incorporate the `Entropy × Prior Ability` interaction term so concise expert queries are not misclassified as over-

reliance.

- **NLP Classifier Validation:** Query intent classification for R_{concept} is validated on ≥ 100 sample prompts with double annotation by 2 independent raters, requiring Cohen's Kappa $\kappa \geq 0.80$.
- **Ethical Masking (Anti-Pygmalion Bias):** Individual student ORI scores and personas are blind-masked from course lecturers during Phase 1 to prevent subjective grading bias. Live lecturer dashboards unlock in Phase 2 post-validation.
- **Protocol Integrity:** All statistical variables, models, and significance thresholds are predetermined prior to data extraction to ensure methodological consistency.

8. Implementation Timeline & Next Steps

Period	Research Activities	Key Deliverables
Weeks 1 - 2	Instrument Configuration, Student Consent, Baseline Phase (Modules 1-2)	Baseline telemetry logs, verified logging pipeline
Weeks 3 - 4	Intervention Phase (Modules 3-4 with active 60s productive friction)	Complete pre/post telemetry dataset
Weeks 5 - 6	MySQL ETL extraction, NLP double annotation ($\kappa \geq 0.80$), EFA factor structure	Psychometric factor structure report
Weeks 7 - 8	Manuscript drafting and submission to IEEE Conference (IEEE TALE / EDUCON)	Submitted conference paper
Semester 2	Phase 2 independent replication (CFA, Hou et al. validation, Scopus Q1 submission)	Scopus Q1 Journal (IEEE TLT / C&E: AI)

Scope Boundary

The ORI metric is specifically tailored for programming assignments with deterministic unit testing in Computer Science. Generalization to non-computing fields is noted as future work.

Action Items for Advisor

1. Deploy S-SPARC in 1 active lab class (35-50 students).
2. Lab session schedule and instructor coordination.
3. Joint review and co-authorship on resulting publications.

References

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